Mega Gate & Mega mHC

MoE routing and mHC residual mixing, one kernel each

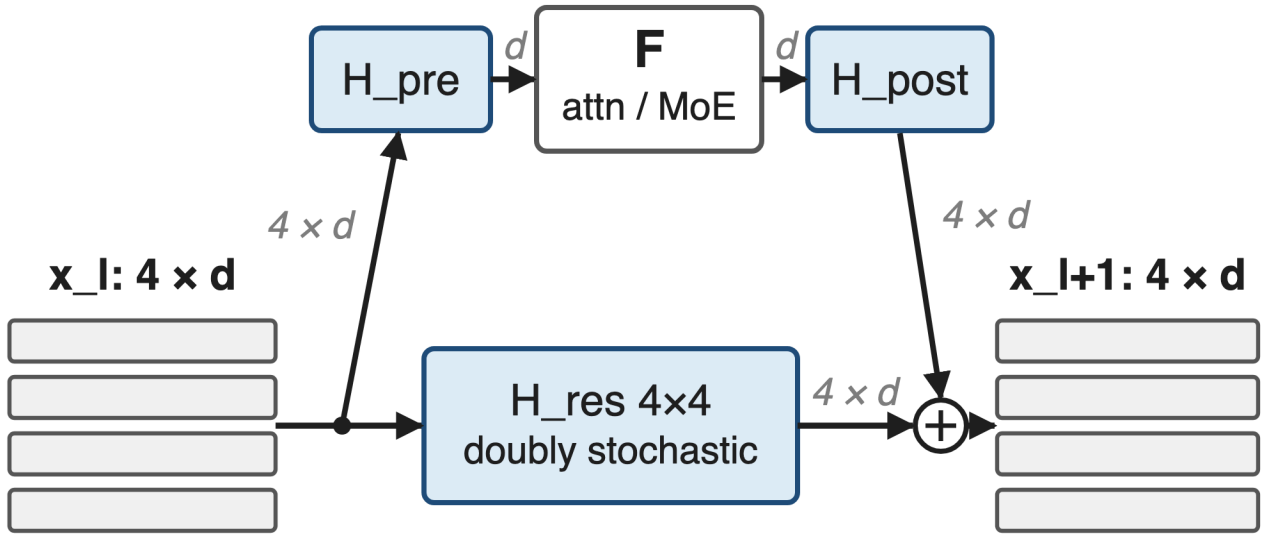
DeepSeek-AI · DeepGEMM (2026-09) · DeepSeek-V4

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Kernel Fusion & Runtime · github.com/deepseek-ai/DeepGEMM

V4 widens every residual stream to four rows

- n = **4** streams: residual is 4 × d per token
- **24** mixing coefficients per token and layer
- Router: √softplus scores, top-**6** of **256** / **384**



24 coefficients per token and layer, from x_l:
RMSNorm · [4d → 24] GEMM · σ and Sinkhorn

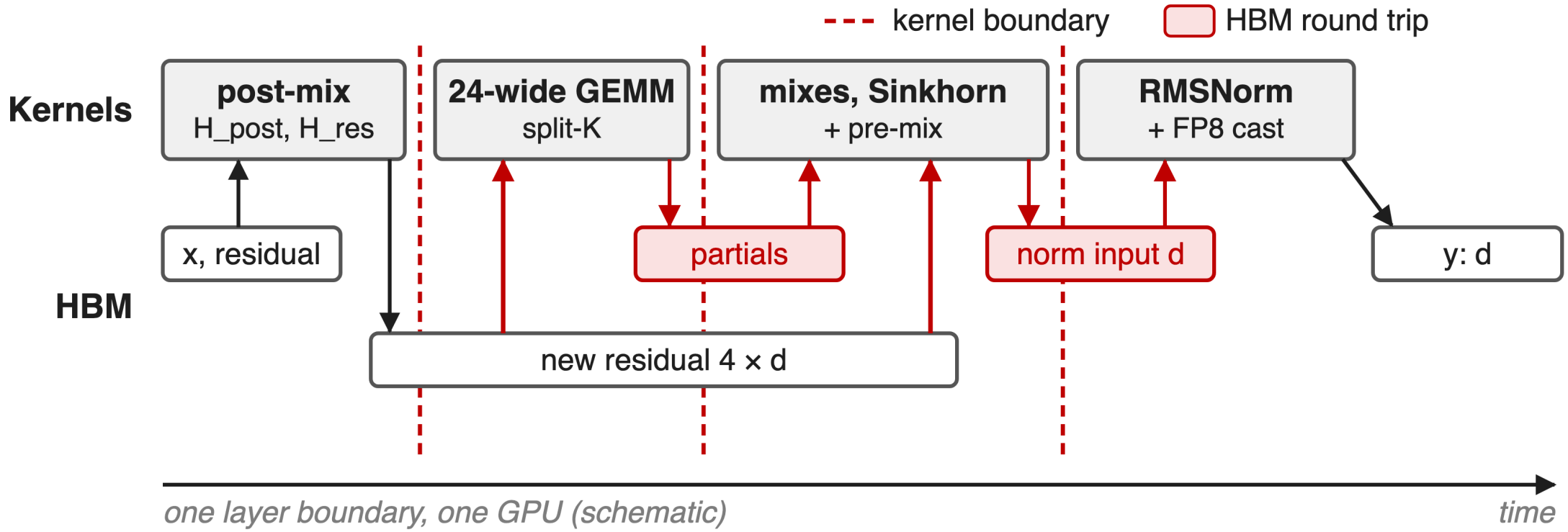
Hyper-connections move about 11× the bytes

≈ 11×

memory traffic per token per layer for 4-stream hyper-connections run op by op: 34d vs 3d elements for a plain residual

Derived from the mHC paper (arXiv 2512.24880), Table 2 at n = 4; excludes the layer's own I/O

Four kernels re-read the stream at every boundary



The new 4 × d residual is re-read twice, for 24 numbers of math

Earlier fusion cut bytes but kept kernel boundaries

Op by op (eager)

- Every mapping and merge is a kernel
- **34d** per token per layer

TileLang kernels (mHC paper)

- Merge fused: (3n+1)d → (n+1)d reads
- **6.7%** training overhead, yet **4–7 kernels** per boundary

Needed: one pass over each token's stream, in one kernel

Insight: everything between two layers is row-local

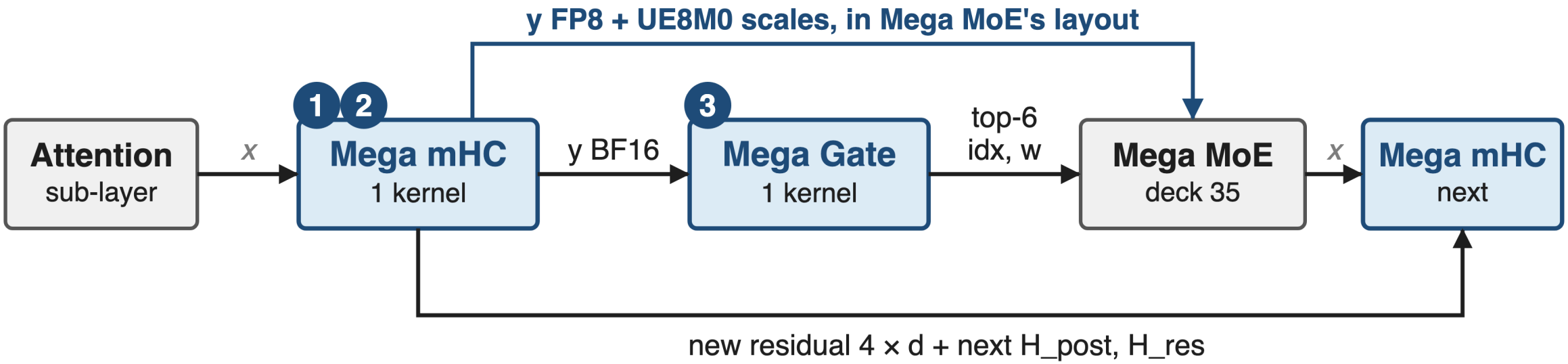
- Post-mix, 24 outputs, Sinkhorn, pre-mix, RMSNorm: one token's row in, one row out
- Only the 4d-long dot products need other SMs (split-K)

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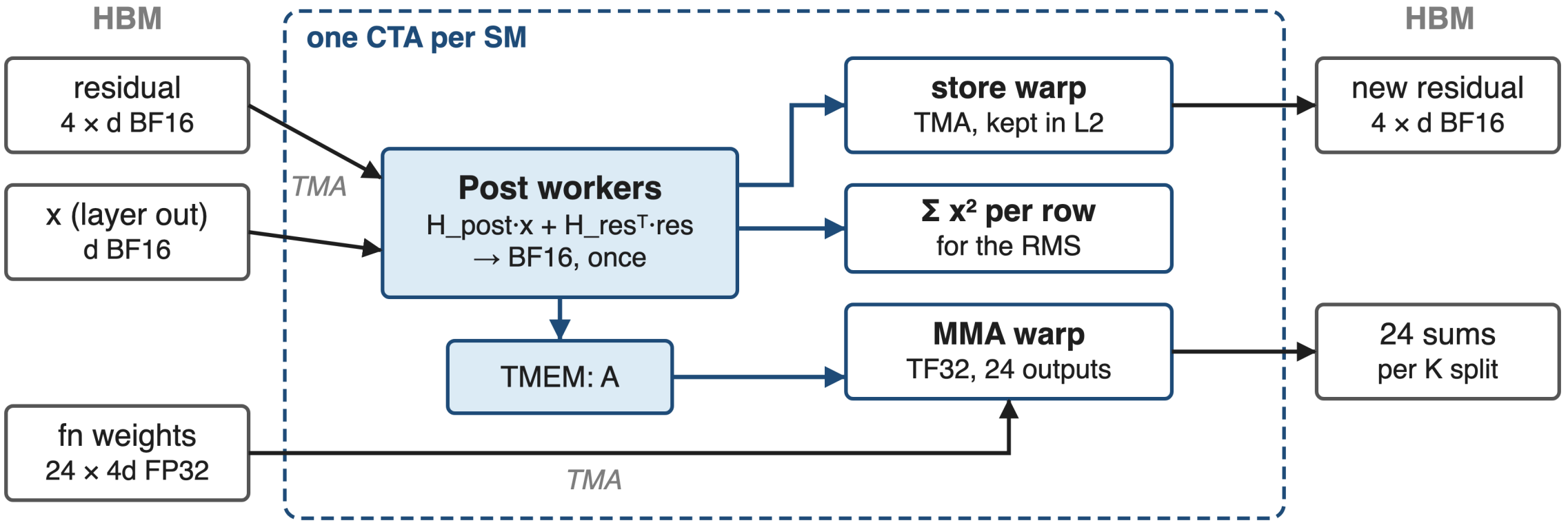
So one kernel can stream each token's residual through once; a router's top-k is row-local too

Both kernels sit around every MoE layer



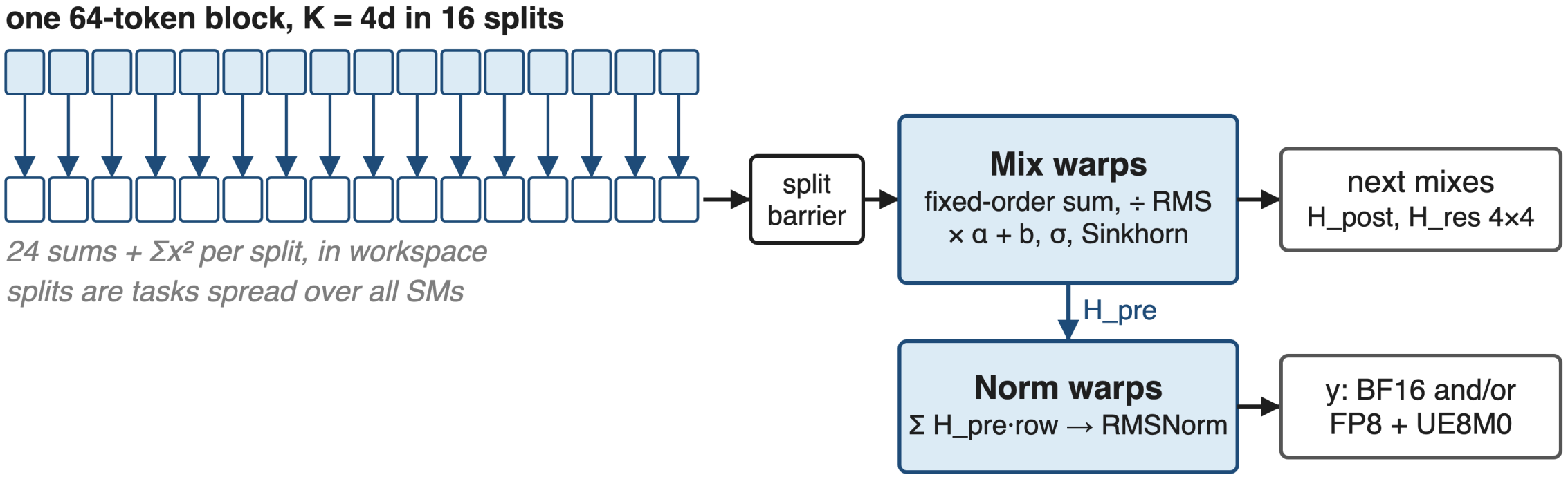
Tensor formats from the DeepGEMM APIs. ❶ ❷ zoom into Mega mHC, ❸ into Mega Gate

❶ One pass feeds store, MMA and RMS



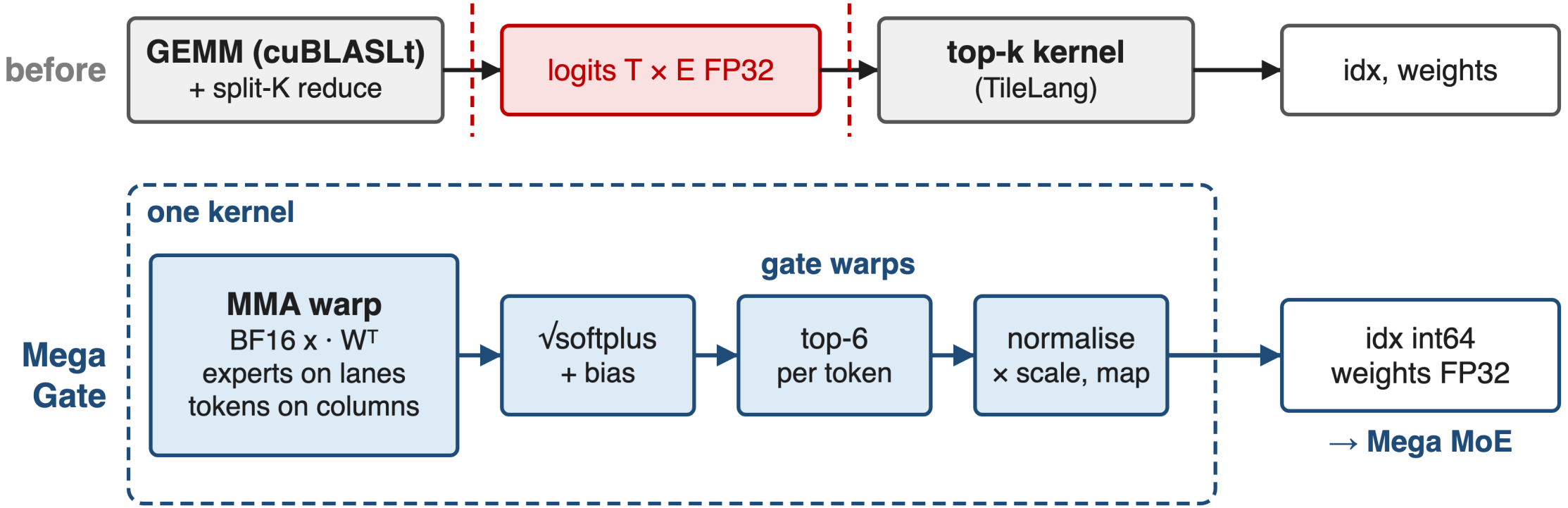
Each new-residual element is computed once; the 24-wide GEMM never re-reads it from HBM

❷ Split-K partials finish inside the kernel



Deterministic mode fixes 16 splits and the summation order: bitwise-stable output

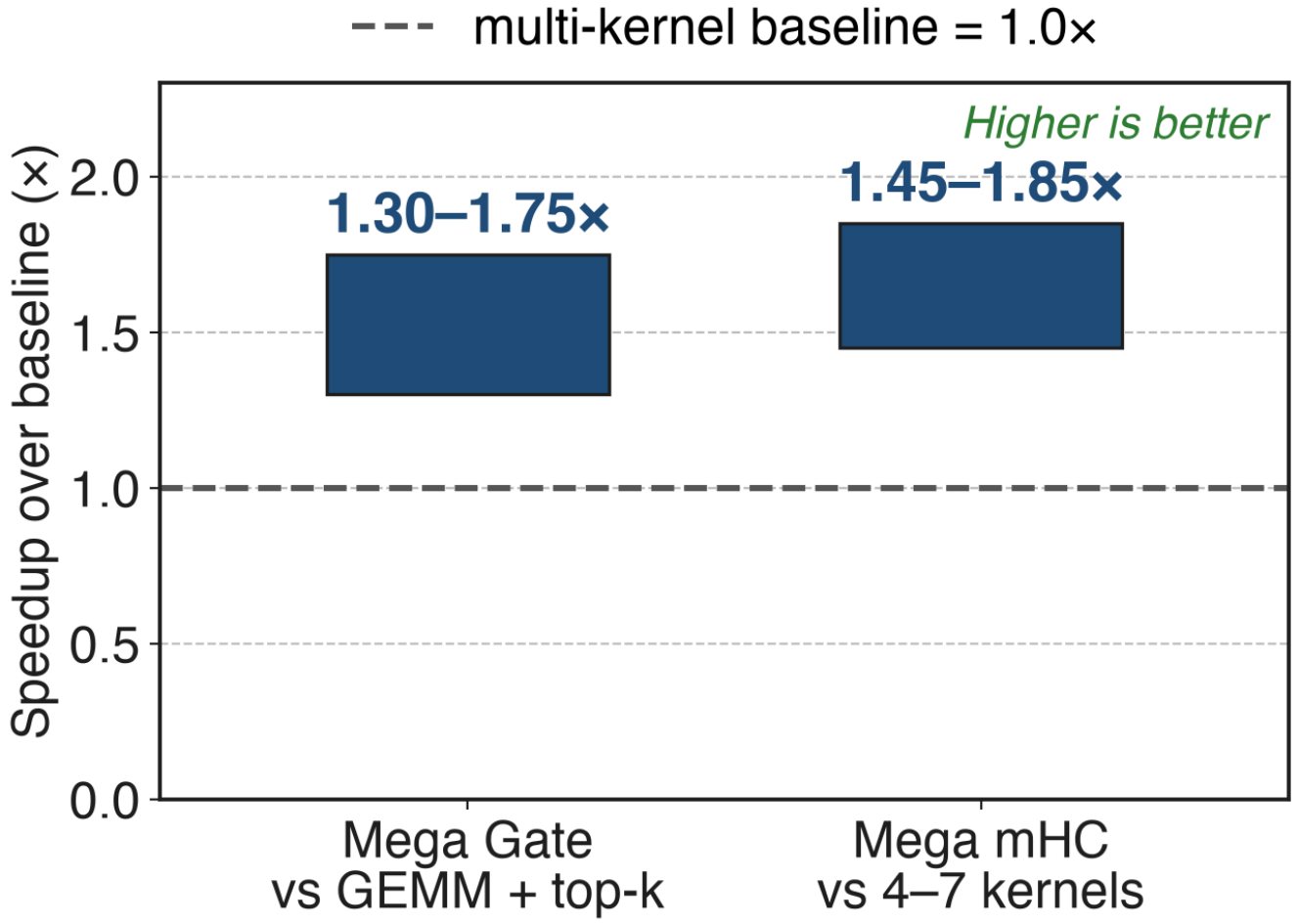
❸ Mega Gate: GEMM to top-6 in one kernel



Ranks on score + bias; emits unbiased scores, normalised over the top-6 and scaled

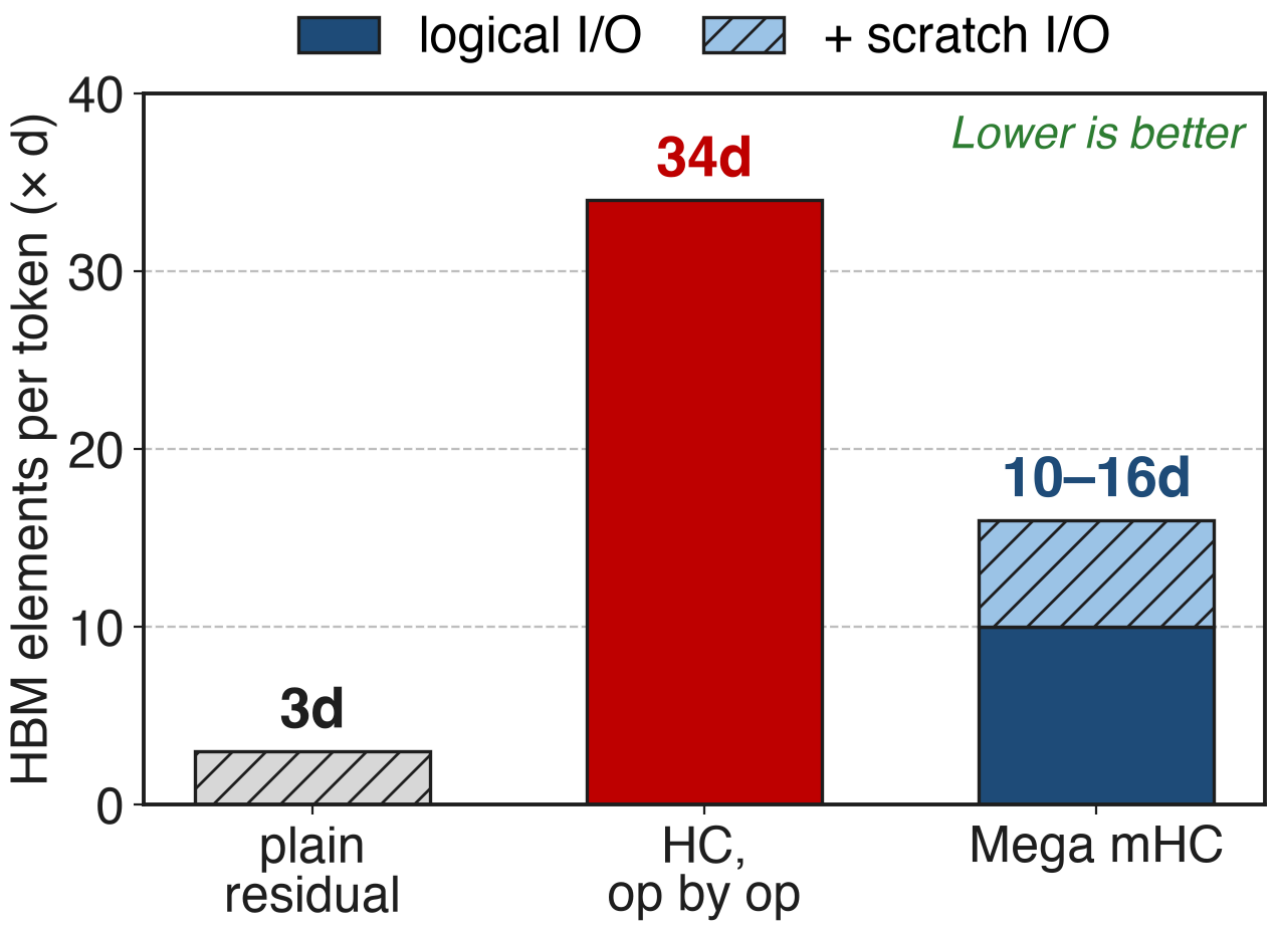
Mega mHC gains 45–85%, Mega Gate 30–75%

- SM100; ranges as published in PR #432
- mHC baseline: TileLang kernels + DeepGEMM GEMM
- Gate baseline: cuBLASLt GEMM + top-k kernel



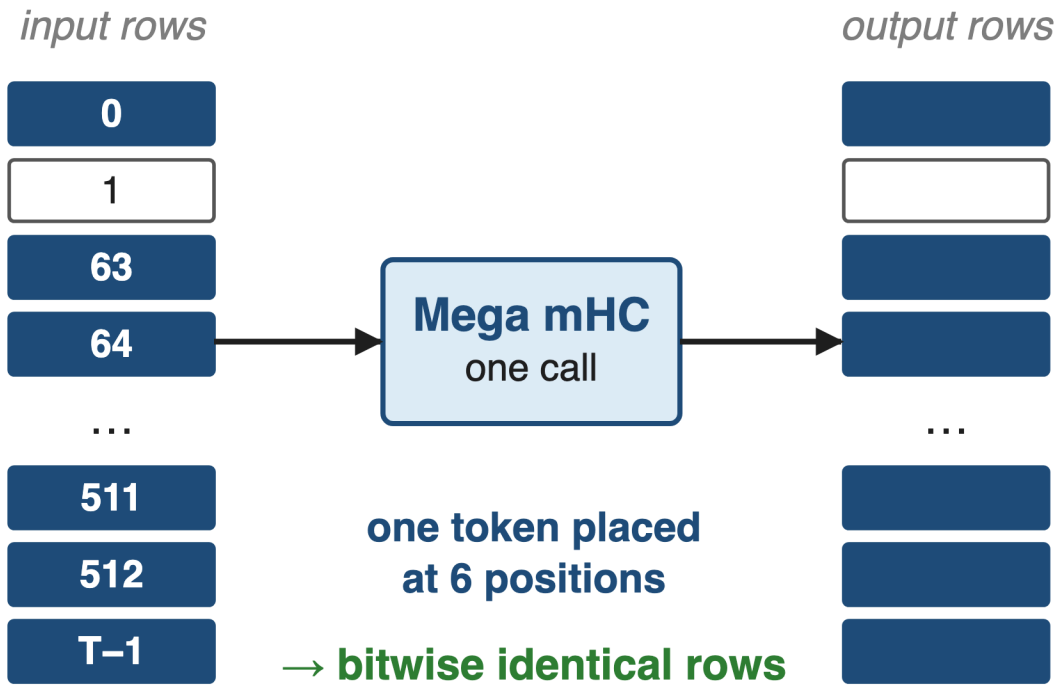
One pass cuts boundary traffic from 34d to 10–16d

- Analytic, n = **4**, per token per layer
- Mega mHC also emits the normalised next input
- Scratch: norm boundary + one L2-kept re-read



Same token, same bits, any batch size

- 30 repeated runs: bitwise equal
- Duplicated tokens: identical rows
- CUDA-graph replay and two streams match eager runs



V4 once needed a second kernel for mHC's deterministic split-K; Mega mHC reduces in-kernel

The cost: SM100, four streams, forward only

Mega mHC

- n fixed at **4**; d a multiple of **1024**
- Forward only; warm streams before graph capture
- New residual still written; normal mode re-reads it

Mega Gate

- ≤ **512** experts, top-k ≤ **32**
- √softplus scoring only
- Both kernels: SM100 only

Summary

- Op-by-op HC: ≈ **11×** a plain residual's traffic
- **Insight**: work between layers is row-local, so stream each row once
- Mega mHC **45–85%**, Mega Gate **30–75%** faster than multi-kernel paths
- Analytic traffic **34d** → **10–16d** per token

https://github.com/deepseek-ai/DeepGEMM